

Inputs (Sec. 4)

TMYx weather: Riyadh, Jeddah
floor plan: 132 zones, 60 units,
9 meters
SBC 602 and pre-code envelopes
unit catalogue; SASO 2663 COP
two-tier volume tariff
comfort band [22, 24] °C

1 Room-resolved model (Sec. 5)

two-node rooms coupled through
walls, slabs and open doors
heat pumps: thermostat, modes,
restart delay, cycling losses
latent load; per-meter bill

2 Formal problem (Sec. 5)

multi-mode system: timed actions,
leaps, infinite schedules,
limit-average cost
minimise the annual bill
s.t. rooms in [20, 24] °C
policy: cooling setpoint, pre-cooling

3 Theory (Sec. 6)

optimal leap: highest level,
lowest mode (or two-mode mix)
infinite-horizon optimum
pre-cooling bound
sources of saving
→ conditions to check

4 Optimisation, evaluation (Sec. 7)

Alg. 2: optimal control (LP)
of the idealised model
Alg. 1: annual simulation,
per-meter bill
49 thermostat settings compared:
select on 60 days, evaluate on 365

5 Verification & validation (Sec. 8)

time-step convergence
heat balance of passive rooms
BESTEST-style 600/900/640/940
EnergyPlus on the same storeys,
with and without setbacks
stock energy use (EUI)

6 Experiments (Sec. 9)

one lever at a time: setpoint,
mode, envelope, equipment,
scheduling
3 cases × 2 cities × 2 seasons
sensitivity analysis

Outputs (Secs. 9–11)

saving of each lever by case,
city and season
SAR a year per apartment
comfort in rooms without units
which lever pays under a
volume tariff

policy θ and constraints (C1)–(C5)

conditions checked